

P (r)

$$P(W) = 0.7$$

$$P(P|W) = 0.9$$

$$P(L) = 1 - 0.7 = 0.3$$

$$P(P|L) = 0.6$$

$$P(W|P) = \frac{P(P|W) P(W)}{P(P)}$$

$$P(P) = P(P|W) P(W) + P(P|L) P(L)$$

$$0.9 \times 0.7 + 0.6 \times 0.3$$

$$0.63 + 0.18 = 0.81$$

$$P(W|P) = \frac{0.9 \times 0.7}{0.81} = 0.778$$

P (2)

$$P(F) = 0.3$$

$$P(R) = 1 - 0.3 = 0.7$$

$$P(D/R) = 0.1$$

$$P(D/RF) = 0.8$$

$$P(F/D) = \frac{P(D/F) P(F)}{P(D)}$$

$$P(D) = P(D/F) P(F) + P(D/R) P(R)$$

$$0.8 \times 0.3 + 0.1 \times 0.7 = 0.31$$

$$P(F/D) = \frac{0.8 \times 0.3}{0.31} = 0.774$$

P(3)

$$P(G) = 0.1$$

$$P(C) = 0.3$$

$$P(N) = 0.6$$

$$P(+|G) = 0.8$$

$$P(+|C) = 0.4$$

$$P(+|N) = 0.2$$

$$P(+) = P(+|G)P(G) + P(+|N)P(N) + P(+|C)P(C)$$

$$= 0.1 \times 0.8 + 0.3 \times 0.4 + 0.6 \times 0.2$$

$$= 0.08 + 0.12 + 0.12 = 0.32$$

$$P(G|+) = \frac{0.1 \times 0.8}{0.32} = 0.25$$

$$P(G|+) = \frac{P(+|G)P(G)}{P(+)}$$

P④

M - Patient has Meningitis

$$P(M) = 0.05$$

$$P(+/M) = 0.95$$

$$P(-/\neg M) = 0.70$$

$$P(+/\neg M) = 1 - 0.70 = 0.3$$

$$P(-/M) = 1 - 0.95 = 0.05$$

$$P(\neg M) = 1 - 0.05 = 0.95$$

$$P(+/\neg M) = 1 - 0.95 = 0.3$$

$$P(-/\neg M) = 1 - 0.3 = 0.7$$

$$\textcircled{1} \quad P(M/+) = \frac{P(+/M) P(M)}{P(+)}$$

$$P(+) = P(+/M) P(M) + P(+/\neg M) P(\neg M)$$

$$0.95 \times 0.05 + 0.3 \times 0.95 = 0.3325$$

$$P(M/+) = \frac{0.95 \times 0.05}{0.3325} = 0.142$$

$$P(M) = 0.142$$

$$P(\neg M/+) = 1 - 0.142 = 0.858$$

$$P(\neg M) = 0.858$$

Not have to disease

$$\textcircled{2} \quad P(M/-) = \frac{P(-/M) P(M)}{P(-)}$$

$$P(-) = P(-/M) P(M) + P(-/\neg M) P(\neg M)$$

$$0.05 \times 0.142 + 0.7 \times 0.858 = 0.607$$

$$P(M/-) = \frac{0.05 \times 0.142}{0.607} = 0.01169$$

$$P(\neg M/-) = 1 - 0.01169 = 0.98831$$

$$P(M) = 0.01169$$

$$P(\neg M) = 0.98831$$

P(-)

$$P(R) = 0.8$$

$$P(+|R) = 0.75$$

$$P(+|\neg R) = 0.15$$

$$P(\neg R) = 1 - 0.8 = 0.2$$

$$P(-|R) = 1 - 0.75 = 0.25$$

$$P(-|\neg R) = 1 - 0.15 = 0.85$$

$$P(\neg R|-) = \frac{P(-|\neg R) P(\neg R)}{P(-)}$$

$$P(-) = P(-|\neg R) P(\neg R) + P(-|R) P(R)$$

$$0.85 \times 0.2 + 0.25 \times 0.8 = 0.37$$

-ve / No Rain

$$P(\neg R|-) = \frac{0.85 \times 0.2}{0.37} = 0.459$$

-ve / Rain

$$P(R|-) = 1 - 0.459 = 0.541$$

P(6)

$$P(c) = 0.0001$$

$$P(J/c) = 0.64$$

$$P(J/\neg c) = 0.6$$

$$P(+/c) = 0.99$$

$$P(+/\neg c) = 0.04$$

$$P(\neg c) = 1 - 0.0001 = 0.9999$$

$$\textcircled{1} P(c/J) = \frac{P(J/c) P(c)}{P(J)}$$

$$P(J) = P(J/c) P(c) + P(J/\neg c) P(\neg c)$$

$$0.64 \times 0.0001 + 0.6 \times 0.9999$$

$$0.000064 + 0.59994 = 0.5999464$$

$$P(c/J) = \frac{0.64 \times 0.0001}{0.5999464} = 0.000106762$$

$$P(\neg c/J) = 1 - 0.0001067 = 0.9998933$$

$$P(c/J,+) = \frac{P(+/c) P(c/J)}{P(+)}$$

$$P(+) = P(+/c) P(c) + P(+/\neg c) P(\neg c)$$

$$= 0.99 \times 0.0001067 + 0.04 \times 0.9998933$$

$$= 0.0001056 + 0.0399957 = 0.040094$$

Date |

Page |

$$P(C/J+) = \frac{0.99 \times 0.000106}{0.040094} = 0.002559$$

(P7) 7.1 Given $P(\text{churn} = \text{Yes} | \text{old}, >2.5 \leq 55)$

Total records = 10
class distribution

Yes = 5, No = 5

$P(\text{Yes}) = 0.5$, $P(\text{No}) = 0.5$

$P(\text{Yes} | X) = P(\text{Yes}) \cdot P(\text{old} | \text{Yes}) \cdot P(>2.5 | \text{Yes}) \cdot P(\leq 55 | \text{Yes})$

$$P(\text{old} | \text{Yes}) = \frac{2}{5} = 0.4$$

$$P(\text{old} | \text{No}) = \frac{0}{5} = 0$$

$$P(>2.5 | \text{Yes}) = \frac{3}{5} = 0.6$$

$$P(>2.5 | \text{No}) = \frac{1}{5} = 0.2$$

$$P(\leq 55 | \text{Yes}) = \frac{3}{5} = 0.6$$

$$P(\leq 55 | \text{No}) = \frac{2}{5} = 0.4$$

for Yes :-

$$\begin{aligned} P(\text{Yes} | X) &= 0.5 \times 0.4 \times 0.6 \times 0.4 \\ &= 0.5 \times 0.144 = 0.072 \end{aligned}$$

for No :-

$$P(\text{No} | X) = 0.5 \times 0 \times 0.2 \times 0.4 = 0$$

The customer is predicted to churn = Yes with probability ≈ 1 (100%)

7.2

without smoothing $\rightarrow$ customer is predicted to churn (Yes) with probability 1

with m - estimate $\rightarrow$ zero probabilities are avoided

Increase m reduce confidence in prediction

Very large m marks classification uncertain

P(8)

Total = 88, P = 84, N = 84

$P(+ | X=7, Y=4)$

$P(\text{Positive}) = 0.5$, $P(\text{Negative}) = 0.5$

for +ve :-

X values : 5, 5, 3, 4

$$\mu_{X+} = \frac{5+5+3+4}{4} = 4.25$$

$$\sigma_{X+} = \sqrt{\frac{(5-4.25)^2 + (5-4.25)^2 + (3-4.25)^2 + (4-4.25)^2}{4}}$$

$$\approx 0.829$$

Y values : 6, 7, 6, 5

$$\mu_{Y+} = \frac{6+7+6+5}{4} = 6$$

$$\sigma_{Y+} = \sqrt{\frac{0+1+0+1}{4}} = 0.707$$

for -ve :-

X values : 1, 3, 2, 8

$$\mu_{X-} = \frac{14}{4} = 3.5$$

$$\sigma_{X-} = \sqrt{\frac{6.25 + 0.25 + 2.25 + 20.25}{4}} \approx 2.693$$

Y values : 7, 4, 6, 1

$$\mu_{Y-} = \frac{18}{4} = 4.5$$

$$\sigma_{Y-} = \sqrt{\frac{6.25 + 0.25 + 2.25 + 12.25}{4}} = 2.291$$

using Gaussian Formula

$$N(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

for +ve :-

$$P(X=7/+) = \frac{1}{\sqrt{2\pi}(0.829)} e^{-\frac{(7-4.25)^2}{2(0.829)^2}} \approx 0.002$$

$$P(Y=4/+)=\frac{1}{\sqrt{2\pi}(0.707)}e^{-\frac{(4-6)^2}{2(0.707)^2}}\approx 0.010$$

$$P(+|x,y)=0.5\times 0.002\times 0.010=0.00001$$

$$P(\text{Positive})=0.00001$$

P(9)

Given $D = \{2.8, 4.2, 5.3, 5.5, 2.1\}$

Mixture of 2 Gaussians:

$$\mu_1 = 2.3, \sigma_1 = 1.8$$

$$\mu_2 = 6.8, \sigma_2 = 2.2$$

$$P(C_1) = P(C_2) = 0.5$$

9.1 log-likelihood

$$P(x) = 0.5 \times N(x; \mu_1, \sigma_1) + 0.5 \times N(x; \mu_2, \sigma_2)$$

* For $x = 2.8$

$$N_1 \approx 0.213$$

$$N_2 \approx 0.033$$

$$P(2.8) = 0.5(0.213 + 0.033) = 0.123$$

$$\log P(2.8) \approx -2.095$$

* For $x = 4.2$

$$P(4.2) \approx 0.135, \log \approx -2.002$$

* For $x = 5.3$

$$P(5.3) = 0.130, \log \approx -2.040$$

* For $x = 5.5$

$$P(5.5) = 0.127, \log \approx -2.065$$

* For $x = 2.1$

$$P(2.1) = 0.123, \log \approx -2.095$$

$$\log \text{ likelihood} = \sum \log P(x_i)$$

$$= -2.095 - 2.002 - 2.040 - 2.065 - 2.095$$

$$\approx -10.30$$

9.2

Yes, we can improve the estimates using the EM algorithm (Expectation - Maximization)

Current parameters are fixed, but data suggests:-

- smaller values (2.1, 2.8) $\rightarrow$ belong to Gaussian 1
- larger values (5.3, 5.5) $\rightarrow$ belong to Gaussian 2
- 4.2 is in between (shared)

New parameters are computed by

- Assigning probabilities (E-step)
- Updating mean & variance (M-step)

This lead to

Better fit to data

Higher log-likelihood